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State-sector shrinkage and workforce reduction in China

Mark W. Frazier

Lawrence University, P.O. Box 599, Appleton, Wisconsin 54912-0599, USA

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Abstract

Restructuring of the state sector in China in the late 1990s reduced the workforce by an estimated 30 to 40 million. This paper uses firm-level data from a survey of 800 enterprises in China to test various explanations for job loss and firm exit among state-owned enterprises (SOEs). The results, based on OLS and binary logistic regression models, suggest that small firms (by output and tax remissions) in labor-intensive sectors, and those in coastal Jiangsu province, had greater likelihoods of firm exit and of workforce decline during the late 1990s. Surviving SOEs continued to report only small reductions in payroll size in comparison to firms under other forms of ownership.
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1. Introduction

In the late 1990s, the state sector in China underwent a massive labor force reduction. Official employment data show that state sector employment peaked at about 109.5 million employees in 1995 before falling to 69.2 million at the end of 2002. The number of workers employed by state industrial enterprises (denoting the manufacturing, mining, and utilities sectors) fell from 44.0 million in 1995 to 15.5 million in late 2002, a 65% decline (SSB and MOLSS, 2003: 23). Urban industrial collectives, firms that are often subsidiaries of state enterprises yet separated for administrative and statistical purposes, saw an equally severe decline in employment, from 14.9 million in 1995 to 3.8 million in 2002 (SSB and MOLSS, 2003: 25).

The fall in industrial jobs in the state and collective sectors was only partially offset by increases in the number of industrial workers employed in other ownership sectors. As Table 1

* Tel.: +1 920 993 6182; fax: +1 920 832 6962.
E-mail address: mark.frazier@lawrence.edu.

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Table 1
Employment changes in industrial enterprises, 1995–2002 (millions)

Ownership form	1995	2002	Net employment change	% Employment change, 1994–1999
State	44.0	15.5	−28.5	−64.8
Collective	14.9	3.8	−11.1	−74.5
"Other"	7.1	18.0	10.9	153.5
Private	3.5	6.4*	2.9	82.9

Sources: SSB and MOLSS 2000, 2003.
* 2000 data.

shows, industrial employment rose from 7.1 to 18.0 million between 1995 and 2002 in a category of ownership that includes state enterprises converted into shareholding and other corporate entities, as well as foreign-invested (including Hong Kong, Taiwan, and Macao) enterprises. In private-sector industry, the number of employees rose from 3.5 million in 1995 to 6.4 million in 2000 (SSB and MOLSS, 2001: 27). Thus, the 13.8 million workers added to payrolls of private and "other" (corporatized and foreign-invested) industrial firms did not compensate for the 39.6 million jobs lost in industrial firms of the state and collective sectors. Even if we include government and service sectors to these job figures, the net job loss is about 23.5 million.[1] State enterprises shed 40.3 million jobs between 1995 and 2002, while the firms in the "other" and private sector categories increased their payrolls by 16.8 million.

This politically sensitive policy of state sector job destruction took place as state-owned enterprises (SOEs) were privatized, sold off, shut down, and converted to shareholding entities (Cao, Qian, Weingast, 1999; Cai, 2002a). The number of industrial state enterprises fell from 110,000 in late 1997 to 53,489 by late 2000, a precipitous decline of 54.7% in just three years. Central government leaders endorsed in the late 1990s a policy to shed small firms from the state sector and to retain partial or full state ownership in the large-scale, heavily capitalized state firms. In 1997, this policy of transforming the state sector based on the scale of a firm was given voice in the slogan, "retain the large, release the small" (*zhuada fangxiao*) (McNally and Lee, 1998). Thus, the decline in state sector employment and the "release of the small" were major contributing factors to the decline in employment in the state sector. Local governments, having received tacit approval from the center, engaged in a wave of mass privatization in which they sold off or shut down small state firms (Cai, 2002a).

Given the disappearance of both state firms and the number of state employees in the late 1990s, it is likely that much of the job destruction in China's state sector was accomplished through either firm exit or conversion of state firms to new ownership forms. This observation raises several important questions related to the sources of job destruction and creation among Chinese firms. Do constraints still exist on the ability of state-owned firms to reduce excess labor, and if so, why? Do SOEs that underwent conversion to shareholding and other corporate forms enjoy more flexibility in making adjustments to the labor force? If state sector workforce reduction was achieved through a policy of firm exit, what were the traits of closed down SOEs versus those allowed to continue operations? More generally, what was the economic and political rationale behind the shrinking of the state sector?

[1] As a result, the number of jobless in China has soared since the mid-1990s. While it is very difficult to estimate the numbers of jobless, estimates cluster around 30 million, a figure that includes those classified as "unemployed," "laid off," and an even more difficult to measure category of jobless rural migrants in urban areas. (Solinger, 2001, 2003) The Chinese government estimated that state-owned enterprises had shed 11.7 million workers in 1999 alone (Xinhua, 2000), and that about half of these were successfully finding re-employment. (Xinhua, 1999).

The existing literature on enterprise restructuring in China and other transition or post-socialist economies suggests several hypotheses that account for the timing and pace of job destruction in the state sector. One explanation stresses the gradual movement of state firms toward more flexible labor practices of those found in foreign-invested and private enterprises. In China, the pattern of reform has been to allow foreign-invested and private enterprises to emerge under more flexible labor market conditions, while the state sector remained relatively insulated from pressures to reduce the labor force. China's reforms and opening to foreign investment created a dual and highly distinct labor policy in which more flexible, market-based labor management practices in non-state enterprises existed alongside a largely unreformed state sector. In the latter, employees could still rely on relatively firm assurances of lifetime employment and extensive benefits that included housing, medical care, and retirement pensions. Labor contracts, introduced in 1986 regulations, were the first attempt to undermine the institution of lifetime employment by placing workers on limited terms of employment, usually of three years. Only by the early 1990s, after nearly a decade of reform in the industrial sector, did state enterprises begin to invoke layoffs and other measures to make them more competitive with non-state firms (Naughton, 1996; Gu, 2001). The 1995 Labor Law established an important precedent by stipulating rights and protections for workers in firms regardless of their form of ownership (Gallagher, 2002). If legal and administrative regulations were becoming standardized across different ownership categories, we should observe state sector firms converging with the flexible employment practices found in firms under alternative forms of ownership in the more competitive non-state sectors. Thus, what might be called the "ownership-convergence hypothesis" would predict that state sector firms would gradually but significantly liberalize employment policies so as to attain autonomy to fire and recruit labor according to market conditions.

An alternative hypothesis also emphasizes ownership forms but predicts divergence rather than convergence. Under this view, state enterprises continue to be constrained in their ability to adopt flexible labor practices, and they retain many workplace institutions of the pre-reform era. (Dong and Putterman, 2001) Other forms of ownership, such as corporatized SOEs and foreign-invested enterprises, have greater leeway in their ability to hire and fire labor. The 1994 Company Law codified several alternatives to state ownership, and many state firms converted themselves to limited liability firms and joint stock corporations with boards of directors and shareholder governance institutions. Ownership conversion, such as taking on foreign investment or adopting a corporate organizational form involving shareholders, allows managers to discard many constraints imposed under state enterprise workplace institutions. (Weller and Li, 2000; Ding and Warner, 2002) The ownership divergence hypothesis thus predicts job losses in transformed and foreign-invested SOEs, but would have more difficulty explaining job losses in traditional SOEs and collectives.

A third hypothesis to account for the pattern of job destruction and creation in China stresses variation in the labor intensity of different industries. Here it might be argued that if one controls for ownership, then job destruction would be more prevalent within low-wage sectors marked by highly competitive product markets, where firms could quickly reduce workforce size in response to market conditions. This hypothesis predicts that state sector firms in the textile industry, for example, might undergo a more rapid pace of job destruction than would firms in more capital-intensive sectors or in those that face less competitive product markets. To the extent that SOEs are found in capital-intensive, "commanding heights" sectors of industry such as steel, energy, defense, etc., this hypothesis is compatible with explanations that emphasize ownership forms, but the source of job destruction has more to do with market structure and competition than with politically motivated protections of workers in all state industrial firms.

A fourth hypothesis stresses the political consequences of job destruction, particularly the threat that unemployment poses to the continuance of Chinese Communist Party rule. China's state sector employees have long been accustomed to the "iron rice bowl" of lifetime employment and extensive welfare benefits delivered by the firm. Turning tens of millions of workers out of state sector employment poses enormous adjustment costs, as well as potential organized opposition to CCP rule. According to this hypothesis, the threat of massive labor unrest accompanying widespread layoffs in the state sector explains the long-delayed dismantling of the iron rice bowl and the reluctance to shut down large, unprofitable state firms prior to the late 1990s. Numerous but locally concentrated labor protests took place throughout China in the late 1990s, as state sector reforms accelerated. (Chen, 2000) The institutional mechanisms devised by the CCP to avoid open unemployment lend support to this hypothesis: workers in state firms are classified as "laid off," "not at post," and "internally retired," rather than "unemployed" or "dismissed." Other regulatory devices allow state firms to cut their payrolls while continuing to remit benefit payments to redundant workers. This policy led to the unintended consequence of laid-off workers engaging in far more numerous protests than the formally unemployed (Cai, 2002b), as those who maintain ties to their firms have a legitimate grievance when employers fail to deliver benefits. Nonetheless, if the political stability hypothesis were taken to its logical conclusion, it is difficult to account for the rapid decline in state sector employment since 1995. If the CCP were obsessed with the threat of labor unrest, it would have proceeded more cautiously than appears to be the case from official employment data. In survey research findings on corporate restructuring and privatization of state-owned enterprises, Oi (2005) has found that the majority of restructuring cases take place at the initiative of local governments. Officials face financial and political constraints in restructuring, and the mix of strategies that they adopt is intended to reduce losses from ailing state firms while minimizing hostile reaction from workers whose jobs are threatened by restructuring.

While each of the above hypotheses is plausible and based upon claims found in discussions of state enterprise reform, few studies have examined firm-level data to account for workforce reduction in the state sector. This article will test the above hypotheses by examining data from a year 2000 survey of 800 firms in China.

2. Enterprise surveys

Data on firms were gathered by a team of researchers under the guidance of scholars at the Economics Institute of the Chinese Academy of Social Sciences in a 2000 survey of 800 state and non-state enterprises in four Chinese provinces and 20 cities. The 2000 sample was drawn from an initial set of 680 state enterprises first surveyed in 1990 and re-surveyed in 1995. When survey administrators attempted to locate the original 680 firms, it turned out that 236 of them did not survive the restructuring of the late 1990s. The 444 surviving firms were surveyed in 2000, and 11 of these had changed ownership forms between 1995 and 2000. Thus, there were 433 SOEs in the 2000 dataset. Surveyors also collected data on 232 urban collectives, 92 corporatized firms, 33 private and joint venture firms, and 10 firms that fit into none of the above ownership categories. A look at employment and other traits of firms from both groups, including the 236 exiting firms and the 433 remaining SOEs, allows us to compare how many jobs were lost through intra-firm workforce reductions and how many jobs were lost through the elimination of an SOE altogether.

Table 2 presents in summary form data on aggregate employment changes between 1994 and 1999 among firms grouped by ownership form. It also presents employment levels at year-end 1994 for the 236 exited firms. The 433 state firms saw their combined workforces fall from 1.11

Table 2
Employment changes, 1994–99, by ownership form

	Ownership form	*N*	1994 Employment (year-end)	1999 Employment (year-end)	Net Employment change	% Employment change, 1994–99
Surveyed firms, 2000	State	433	1,112,417	1,022,355	−90,062	−8.1
	Collective	232	277,373	263,710	−13,663	−4.9
	Joint stock	17	5995	5918	−77	−1.3
	Limited liability	75	112,339	103,900	−8439	−7.5
	Private	12	11,423	10,750	−673	−5.9
	Joint venture	21	25,837	23,204	−2633	−10.2
	Other	10	6861	6441	−420	−6.1
	Subtotal	800	1,552,245	1,436,278	−115,967	−7.5
Surveyed firms, 1995	State, surviving firms*	444	1,137,976	1,022,355	−115,621	−10.2
	State, exited firms, 1994–1999	236	212,552	n/a	−212,552	n/a
	Subtotal	680	1,350,528	1,022,355	−328,173	−24.3

* Includes 433 state firms and 11 state firms that would convert to alternative ownership form (e.g., limited liability) in 1994–1999.

million at year-end 1994 to 1.02 million at the end of 1999, an aggregate decline of 8.1%. In the other ownership sectors, some firms had far more rapid employment declines and some registered significant job gains in 1994–99. Calculated by firm-level means (not shown in Table 2) the average proportional change in employment by ownership categories was as follows: SOEs fell by 8.3%, collectives rose by 1.09%, joint stock firms rose by 8.65%, limited liability firms fell by 3.2%, and private enterprises fell by 7.18%. High standard deviations suggest a great deal of variation within these ownership categories.[2] The overall job destruction rate among the firms surveyed in 2000 is not what one would expect, given that official PRC statistics reported a 64.7% decline in employment from 1995 to 2002 among state sector industrial firms (defined as manufacturing, mining, and utilities). Between 1995 and late 2000, roughly the same time covered in the surveyed firms, the decline in employment was 52.4%. This glaring difference between employment changes as reported in the official data and those from the enterprise-level contained in the sample can be explained in two ways. First, state-owned enterprises in most surveys, including this one, report the total number of workers and staff as the sum of all those on the payroll, including employees considered "not at post." These employees are eligible for benefits from the firm but are no longer reporting to work or earning wages. The State Statistical Bureau and the Ministry of Labor of Social Security, whose figures are reported in Table 1, report employment in state-owned enterprises as the number of workers considered "at post."

Second, the national statistics on the decline in state sector employment reflect the exit of firms from the state sector, while employment data in the 2000 survey contain "survivorship bias." Firm exit accounts for a substantial portion of the decline in state sector employment, and this is reflected in the line in Table 2 that shows information on 236 exiting firms from the

[2] An analysis of frequencies (not shown in Table 2) reveals the wide dispersion of mean employment changes within individual ownership sectors. Among the 232 collectives, ten percent saw their workforces shrink by 33% or more, while another 10% had expansions of 30% or more. Twenty percent of the 92 corporatized firms also had employment declines and gains of similar or greater magnitude in 1994–99. By contrast, at the "extreme" end of the SOEs in the sample, only five percent of the firms had employment reductions of more than 16.6%.

Table 3
Traits of firms liquidated in late 1990s, versus firms extant in 2000

Status of state enterprise		Average annual wage for production workers	Total workers and staff	Industrial output value (million RMB, 1990 prices)	Gross profits (million RMB)	Annual output per employee (RMB/ worker)
Extant in 1999	Mean	4406	2563	115.64	1.18	49,333
	n	443	444	444	443	444
	(sd)	1896	4104	414.0	11.0	48,914
	Total	1,951,947	1,137,976	51,343.51	523.70	21,903,944
Liquidated after 1994	Mean	3604	901	14.43	−.775	15,077
	n	236	236	235	236	235
	(sd)	1195	1963	66.16	1.71	11,516
	Total	850,533	212,552	3,390.63	−182.80	3,543,032

survey. These firms, all SOEs, reported 212,552 employees at year-end 1994. Almost all of these 212,552 workers lost jobs as their firms were either liquidated or purchased by outside investors in the late 1990s.[3] Assuming that these workers permanently left the state sector, then the percentage decline in state sector employment among the sample can be estimated as follows: employment among the 680 SOEs in 1995 fell from 1,350,528 that year to 1,022,355 (in 433 remaining SOEs) in late 1999, a decline of 24.3%. This figure approximates the Ministry of Labor and Social Security statistics which show a 27% decline in employment in industrial SOEs between 1994 and 1999.[4]

Table 3 shows the striking contrast, as of year-end 1994, between the employment levels, average wages, value of industrial output, productivity, and profitability of the 444 surviving firms and the 236 firms that exited after 1995. This represents an exit rate of 34.7%, which is less than the rate of firm exit reported in national statistics for the late 1990s. As noted above, the national rate of firm exit was 55% between 1997 and 2000 alone. Nonetheless, the data in Table 3 are in line with accounts of firm exit by smaller SOEs. The surviving SOEs in the sample were larger employers, with higher average wages, productivity levels, and profits. The industrial output value of the surviving firms was on average about 8 times greater than that of the exiting firms, and surviving firms reported over three times productivity of exiting firms.

These summary statistics on employment levels in the sample suggest that firm exit may have been a prominent pattern in state sector workforce reduction, but we are still left with several questions: Were state firms from the sample converging toward the flexible labor policies found in the non-state sector? What effects did demand for labor across different industries have on job destruction and creation? To what extent did political constraints

[3] The research team that collected data in the 2000 survey also pursued the question of what happened to the 236 "exit" firms and reported short summaries on the disposition of these firms. From these summaries, it is apparent that 145 out of the 236 firms were either liquidated or went through bankruptcy, or simply ceased operations. The remainder of cases (91) were either combined with other firms or purchased by foreign investors, but it is not clear in these instances how many, if any, employees retained their jobs. While it is not possible to claim that all 212,552 workers out of the 1.35 million employees in the 1995 sample would have immediately lost their jobs when their firms were bought out and converted to foreign or other types of ownership, it is very likely that a large portion of these employees are now out of state sector employment.

[4] Official statistics (SSB and MOLSS, 2001: 22) reported a 45% decline in employment over the late 1994-late 1999 period, which includes only "at post" workers. The Ministry of Labor and Social Security (MOLSS, 2001: 570) estimate of 27% includes in its totals "not at post" as well as laid-off workers.

influence job destruction across different firms? What accounts for firm exit and the accompanying job loss in the state sector?

To address these questions, an ordinary least squares regression model was developed to explain job loss among firms in the late 1990s, and a binary logistic model was utilized to account for firm exit in the late 1990s. In both models, the same explanatory variables were selected in order to see whether the same variables could account for job loss and firm exit.

3. Explanatory variables

The models used three groups of independent variables: 1) those that captured initial conditions of employment and output in the first data year (1994); 2) those that grouped the firms into 8 industrial categories, in order to test the effects of intersectoral differences on workforce reduction and firm exit; and 3) those that served as indicators of political constraints on workforce reduction and firm exit.

The baseline condition variables represent attempts to identify more precisely the underlying dimensions of firm scale. While it is apparent that many firms exited the state sector, what exactly does "small" mean in the policy slogan "release of the small?" Does it refer to numbers of employees, value of output, or something else? If small-scale firms were cut and large-scale firms retained, were the latter necessarily more productive than the former? To address these questions, the logarithms of employment in 1994 and of output value in 1994 prices were included in the regression models.

Two other variables indicating initial conditions were included based on descriptive statistics that showed potentially significant distinctions across ownership form. Among the sample firms, there was a strong divergence between SOE and other forms of ownership in the percentage of workers classified as temporary labor.[5] SOEs in the sample reported having 14.5% of their workforce as temporary labor in 1999, and this figure did not vary significantly with levels prevailing in 1994. In collective, joint stock, limited liability, and other types of firms, it is apparent that temporary labor is the norm. The average portion of the workforce classified as temporary labor ranged from 67% in joint ventures to 87% among collectives. What does it actually mean to be considered temporary as opposed to full-time labor? It is likely the case that the non-state firms designated large majorities of their workforces as temporary in order to avoid Labor Law and social insurance regulations that cover full-time workers but do not, under certain conditions, cover workers classified as temporaries. For example, classification as temporary labor would make it easier for firms to cut back on the labor force in response to market conditions. Thus, the percentage of temporary workers serves as an indicator of flexibility in a firm's labor recruitment practices.

By other measures of workforce composition, such as the percentage of rural labor hired in the 1990s, the portion of women in the workforce, or the respective share of employees classified as production workers versus non-production workers, ownership effects are significant in some cases but not large. For example, the average share of rural workers among new hires in the late 1990s was about 25% for SOEs versus a range of 41% to 49% for collective, joint venture, and private sector firms. SOEs had a comparable share of women as a portion of the total number of employees, at 44%, while the percentage of women in the

[5] This is not, it should be emphasized, synonymous with "contract labor," or employees who have signed labor contracts, usually of 1 to 3 years, with employers. Virtually all employees in the sample firms were on labor contracts by 1999. Temporary labor (*linshi gong*) instead denotes a contrast with "official" or "full-time labor." (*zhengshi gong*).

workforce of collectives and corporatized firms ranged from 41% to 46%. SOEs also had a 58% share of their labor force classified as "front-line production workers," similar to the 57% to 66% range reported by non-state firms.

A second group of variables were used to test the extent to which variation in industry, such as light or heavy, capital or labor intensive, might explain changes in workforce levels. The firms surveyed in 2000 fell into eight industry categories. For the regression analysis, dummy variables were used for each industry category, with textiles removed to represent a baseline category. Average wage levels in 1994 were also included as an explanatory variable to test whether high wage firms, after controlling for the effects of different industries, might be significant in predicting changes in employment among firms.

A third group of variables served as indicators of political constraints on job loss and firm exit. These variables included region, revenues for local governments, and various constraints imposed on management by government agencies and Communist party branches. These variables require some elaboration.

For regions, the regression models used dummy variables for the four provinces. Three of these provinces (Shanxi, Jilin, and Sichuan) are dominated by state industry, whereas Jiangsu province, a coastal province that includes the prosperous Yangzi River Delta, possesses far more diverse ownership forms and is generally regarded as a wellspring of activist, growth-promoting local officials. In the regression, Jiangsu was used as a dummy variable to test whether a firm's location there made a significant difference in changes in workforce levels in the late 1990s.

Central and local governments have long relied on the profits of state firms as sources of revenue; thus, the total taxes remitted by sample firms provide a sound basis for estimating the effects of revenue provision on employment. A political economy explanation for firm change, or firm exit, could be that providers of large revenues would be protected from job cuts or bankruptcy by local officials who benefit from the revenues generated by profitable local firms. Also included in the group of political variables was the administrative level at which a firm is supervised. That is, state firms under the administrative supervision of central government agencies might face different prospects for employment changes and firm exit than would firms regulated by a county or municipal agency. Collectives generally fall under regulatory control of agencies below the county and city levels.

The regression models also include two items from a questionnaire given to factory directors: the extent to which constraints on hiring and firing workers posed a problem in the operation of the firm; and the degree to which the local Communist party exerted influence over personnel appointments in the firm. These questions were administered in 2000, and to the extent that the responses reflect the factory director's perception of political constraints in recent years, they can still be used as a predictor of employment changes and firm exit in the late 1990s.

Enterprise directors were asked to offer their assessment of specific management problems believed to be common among Chinese firms. Among such management problems was the possibility that managers "lacked the authority to hire and fire workers." In the Chinese context, this meant that they had to seek approval from some external authority, such as a local industrial bureau, the firm's board of directors (which might well contain local officials), the local Communist Party committee, etc. Table 4 shows in cross-tabs format the wide disparity in how directors of SOEs, collectives, and limited liability firms (other ownership forms were left out for simplicity's sake and because of their relatively small N) viewed constraints on making labor force adjustments. Slightly over half the respondents reported that a lack of autonomy to hire and fire workers posed a management problem for them. However, the clear contrast lies in ownership, as 83% (316 out of 380) of those claiming some or many difficulties in hiring and

Table 4
Ownership form and hiring autonomy

Ownership form		Does lack of autonomy to recruit and dismiss labor pose a managerial problem?			
		No difficulties	Some difficulties	Many difficulties	Total
SOE	*n*	117	217	99	433
	% Row total	27.0	50.1	22.9	100.0
Collectives	*n*	189	43	0	232
	% Row total	81.5	18.5		100.0
Limited liability	*n*	54	16	5	75
	% Row total	72.0	21.3	6.7	100.0
Totals	*n*	360	276	104	740
	% Row total	48.6	37.3	14.1	100.0

Pearson $X^2 = 206.9$; $p < .001$.

firing workers were from SOEs. Large and significant majorities of collective and limited liability firms reported no difficulties (81% and 72%, respectively).

This contrast in the ability of management to increase or reduce (most likely, the latter) workforce levels goes far in explaining the relatively modest 8.3% mean reduction in employment in the sample SOEs in the late 1990s. Yet it also shows that SOEs, in order to survive, have strong incentives to change their ownership form by re-incorporating as limited liability or joint stock companies. Indeed, among the 367 non-SOE firms in the 2000 survey, 22 had been SOEs in the recent past. As a group, the 22 ex-SOEs had an 11% decline in employment levels over the same period. Of these, 7 had been re-incorporated as limited liability corporations-and their aggregate workforce levels declined by 16.3% in the late 1990s (mean of 15.7%; SD = 18.6). This figure is almost double the 8.3% average workforce decline registered among SOEs in the sample.

The amount of retirees in proportion to current employees constitutes another political variable. This is because state as well as collective and even corporatized firms in the late 1990s either had to support retirees directly by providing them with pensions and medical benefit payments, or they were levied payroll taxes by local governments based on local retiree populations. Moreover, as frequent news reports of pension protests suggested, firms that exited through bankruptcy or other channels often left ex-employees, and their retirees, with no social safety net to speak of. Such protests posed direct and obvious problems to local governments, and were a serious concern among officials charged with maintaining social stability.

The possibility that SOEs used China's reformed pension system as a vehicle through which to reduce their workforces receives considerable support in the survey data. Between 1994 and 1999, a total of 159,810 workers in the sample left SOE employment (not a net figure as reported in Table 1 but an absolute number of separations). Of these, nearly 145,000 (90.4%) were through retirement, as opposed to dismissals, resignations, or transfers. By contrast, out of 72,145 who left collective sector employment, nearly 40,000 (55.3%) did so through dismissal. Roughly the same proportion of those separated from corporatized firms did so through dismissals. It appears that SOEs, unlike non-SOEs, shed workers by retiring them and then relying upon transfers or negotiated arrangements with local governments to offer financial support for the retirees.

This clear contrast in the manner in which state and non-state firms released workers strongly suggests important constraints on the ability of state enterprises to use flexible labor policies to compete domestically and internationally. Instead of adjusting the labor force in response to

market conditions, SOEs channeled their excess workers through retirement and pension procedures. This pattern of job loss through attrition suggests a potential explanatory variable in predicting workforce reduction among SOEs: firms with larger numbers of retirees (proportional to the number of workers on the payroll) would be expected to reduce their workforces more than firms with fewer retirees.

State enterprise managers also reduced their labor forces through a second institutional mechanism, known as lay-offs (*xiagang*). As it moved to shrink the state sector, the Chinese government introduced regulations in the 1990s that allowed firms to cease paying workers whom management could classify as "laid off." These workers were formally on the payroll and eligible for subsistence payments until they found new jobs elsewhere. (Lee, 2000) Laid-off workers (LOWs) are not considered formally released from their enterprises since LOWs maintain "labor relations" with their firms by being eligible for benefits, minimum subsidy payments, and re-employment in the unlikely event that production lines are revived. (Appleton et al., 2002; Cai, 2002b) LOWs are thus considered officially documented employees, and their numbers are included in employment figures when enterprises report total workers and staff. SOEs in the sample reported a total of 28,780 LOWs between 1994 and 1999, a surprisingly low figure in proportion to the approximately 1.05 million employees in the 800 sample firms (about 2.6%). Official statistics reported a total of 9.4 million LOWs for 2000 (SSB and MOLSS, 2001: 401) among an SOE workforce of 78.8 million (about 13%).[6] The share of laid-off workers was thus not included as an explanatory variable in the regression model to predict job loss.

4. Regression analysis: explaining workforce reduction, 1994–1999

OLS regression was used to account for job loss among sample firms, with the dependent variable the change in employment between 1994 and 1999, expressed in logarithmic form.[7] Table 5 displays the effects of explanatory variables upon different ownership forms. These models were chosen based on the initial findings suggesting considerable variation between SOEs and other forms of ownership. Thus, model 1 separately tests the effects of explanatory variables on job losses among the 433 SOEs, and model 2 does the same for the 221 urban collectives. Model 3 presents a simpler analysis, showing the effects of two explanatory variables on job loss among the 134 enterprises of other ownership forms (corporatized SOEs, joint ventures, and private firms.) The explanatory variables in model 3 were pared down considerably because of the smaller sample size. Rather than analyzing the effects of all variables in models 1 and 2, model 3 provides a further test of the most significant finding in models 1 and 2, related to baseline firm employment levels.

Employment in 1994, the baseline year, was a significant explanatory variable for employment change in the late 1990s, across all the ownership forms in Table 5. The coefficient

[6] This discrepancy between LOW statistics nationally and those of the sample suggests a number of potential explanations. First, since the sample firms are all industrial, it could be the case that LOWs are more commonly from non-industrial sectors. Second, it is possible that firms in the sample may have been offering rosy estimates of how many workers were designated as LOWs. However, it can be in a firm's interest to over-report LOWs, since this shows their compliance with national policy and bolsters their arguments to local governments that their firm needs subsidies in order to support LOWs. Perhaps the most likely explanation for the small numbers of LOWs among the 433 state enterprises in the sample is that they were financially healthier than the actual universe of SOEs in China.

[7] The OLS model was also run using the percentage change in employment as the dependent variable, with largely the same results and directions on coefficients.

Table 5
Results of OLS regression

	Model 1 (SOEs) $n=433$ $F=1.80$ sig.$=.019$ $R^2=.080$		Model 2 (collectives) $n=221$ $F=2.25$ sig.$=.003$ $R^2=.136$		Model 3 (corporatized, private, etc.) $n=134$ $F=3.70$ sig.$=.027$ $R^2=.053$	
	Unstd. coefficients	Standard error	Unstd. coefficients	Standard error	Unstd. coefficients	Standard error
(Constant)	−.0410	.0400	**−.5392****	.2291	−.2756	.2091
Log 1994 employment	**.0117****	.0059	**−.1126*****	.0347	**−.0551****	.0246
Log 1994 output value	.0014	.0053	.0382	.0377	–	–
% of temporary workers, 1994	−.0001	.0002	−.0003	.0008	–	–
% of women employees, 1994	.0002	.0001	.0000	.0000	–	–
Log 1994 average wage	−.0004	.0093	.0166	.0522	**.1131***	.0590
CCP influence	.0024	.0026	−.0059	.0194	–	–
Hiring autonomy	−.0003	.0019	−.0043	.0275	–	–
Level of reg. agency	−.0001	.0019	.0048	.0087	–	–
% of retirees	**−.0005****	.0002	**−.0030***	.0016	–	–
Log 1994 taxes	**−.0058****	.0029	**.0971*****	.0311	–	–
Mining dummy	.0021	.0064	−.0283	.0465	–	–
Machinery dummy	**.0077***	.0400	−.0394	.0383	–	–
Utilities dummy	**.0148***	.0060	–	–	–	–
Food dummy	.0007	.0052	−.0242	.0473	–	–
Chemicals dummy	.0030	.0002	.0238	.0340	–	–
Metals dummy	.0075	.0001	.0137	.0363	–	–
Furniture–paper dummy	.0000	.0093	.0133	.0419	–	–
Sichuan dummy	−.0062	.0026	**.0566****	.0264	–	–
Shanxi dummy	**−.0103****	.0019	.0015	.0303	–	–
Jilin dummy	−.0045	.0019	.0052	.0327	–	–

Dependent variable: log employment 1999 minus log employment 1994.
Coefficients for significant presented in boldface.
*$p<.10$; **$p<.05$; ***$p<.01$.

for employment in 1994 is positive for SOEs and negative for collectives and for alternative ownership forms. The coefficient suggests that all else equal, SOEs with large numbers of workers in 1994 saw a slight increase in employment compared to SOEs with much lower 1994 employment levels. For the urban collectives (model 2) facing a more competitive environment, larger employment levels in 1994 were associated with declines in employment during the late 1990s. The other ownership forms (model 3) exhibited a similar tendency for large firms to cut employment to a greater extent than smaller firms. Expressed in percentage terms, the coefficients suggest that for urban collectives, a change in firm size (specifically a firm with 100 compared to 1000 employees) would be associated with a 29.6% decline in employment over the late 1990s. Larger corporatized SOEs, joint venture, and private firms saw a 13.5% decline compared to smaller firms. The opposite results obtained for traditional SOEs in model 1, where larger enterprises were associated with a 2.7% increase in firm payrolls during the late 1990s. This result should be placed in context, however. As noted above, SOEs in this survey and others report all employees on the payroll, even those who are no longer engaged in work at the firm but are only eligible for benefits. It is quite possible that many SOEs in practice cut a comparable percentage of their workforce as did other non-SOE firms. Since the SOEs were still held

responsible for various benefits for these workers, the payroll numbers remained relatively steady.

The other baseline year variables, such as average wages, proportion of temporary workers and women employees had very small and insignificant coefficients in models 1 and 2. This finding further reinforces the point that once the employment level is taken into account, none of these other variables is significant in explaining employment changes in the late 1990s. The notable exception is for firms in model 3—the model in which corporatized SOEs, joint venture, and private firms are grouped together. Here, it is apparent that after controlling for size of employment in 1994, the average wages had a large and significant impact upon employment. The positive coefficient indicates firms with high average wages in 1994 experienced a significant increase (29.7%) in employment over the late 1990s. Because of the small sample size, this model did not include the variables for particular industries, but the wage and employment variables for 1994 in model 3 likely reflect increases in productivity and wages for these firms throughout the 1990s (e.g., fewer workers needed and higher wages paid).

Table 5 also shows that how much a firm remitted in taxes in 1994 was associated with changes in employment over the late 1990s. Here too the signs on coefficients were reversed between SOEs and collectives. The results suggest that urban collectives that paid higher taxes–which can be interpreted to mean that they were more profitable–were associated with substantial gains in employment in the late 1990s. For SOEs, the opposite was true, with firms that paid higher taxes shedding more labor during the late 1990s than firms that paid few or no taxes. This finding might be interpreted as a consequence of more profitable (and therefore more taxable) SOEs in the mid-1990s being granted more leeway to invoke workforce cuts in the late 1990s. For unprofitable SOEs in the mid-1990s, the government response was not to invoke reductions in payrolls but to pursue firm exit (as seen in the exit model below).

The results of models 1 and 2 in Table 5 also support the hypothesis regarding industrial sector as a source of job destruction. Model 1, for SOEs, shows positive coefficients for all industries, signifying that the more competitive textile sector (the baseline sector) was associated with decreases in employment when compared to all other sectors. The coefficient for utilities, a monopolized and more capital-intensive sector than textiles, was significant. For collectives, the results were far less conclusive, with coefficient signs both positive and negative (and none significant) across different sectors compared to textiles. Without knowing more about the relative factor endowments and competitive environment of these firms, it is difficult to reach any conclusions on the effect of sector on employment levels.

The political hypotheses that predicted constraints on shedding labor because of high levels of Communist Party influence, lack of autonomy over employment decisions, or level of the firm's regulatory agency did not yield significant coefficients in this model. SOEs regulated by lower level government agencies cut workforces more than did SOEs under higher level government agencies. Firms that reported many problems arising from a lack of labor flexibility were, ironically, associated with declines in workforce levels. Perhaps their efforts to reduce labor had brought about the problems that enterprise managers expressed in the survey. In an unexpected result, SOEs that reported decreasing levels of Communist Party interference were associated with increases in employment during the 1990s.[8] This result shows that Party interference at the firm level should not necessarily be equated with constraints on labor force reductions.

[8] The variable was coded from 1 (high interference) to 3 (no interference).

Finally, the results point to the significance of region in explaining job destruction and creation. For SOEs, all the signs on the provincial dummies are negative, meaning that SOEs in Sichuan, Jilin, and Shanxi saw relatively smaller declines in their workforces compared to SOEs in Jiangsu. The magnitude of these declines is not large—in model 1, the Shanxi coefficient was significant at the $p < .10$ level (Jilin and Sichuan were at $p < .20$), which indicates that all else equal, an SOE in Shanxi saw a 2.3% decline in employment compared to an SOE in Jiangsu. This finding could be interpreted in one of two ways. First, there is likely to be a locational benefit that Jiangsu SOEs enjoy from their being in a prosperous coastal area with access to the urban markets of Shanghai and other cities in the lower Yangzi River Delta. Jiangsu's per capita GDP in 2000 was about 11,800 yuan, compared to 4800 yuan for Sichuan, 6850 yuan for Jilin, and 5100 yuan for Shanxi. A second possible interpretation is that Jiangsu province, famous for its activist local governments and officials who have promoted rural industrialization and economic growth by offering administrative guidance to local firms, may also have shielded their SOEs from substantial job loss in the late 1990s. Model 2, however, challenges the latter argument by showing that for collectives, the sign on the provincial dummies is reversed. All else equal, collectives in Jiangsu shed labor while firms in the other three provinces did not, or slightly increased labor. (In a preliminary test of the corporatized and private firms in model 3, the provincial dummies were also positive but insignificant, and were removed from the final version of model 3.) These results lend greater support to the explanation that a more competitive market environment in Jiangsu generated greater flexibility in firm labor decisions. If this is true, then the analysis of firm exit should also show that location in Jiangsu was associated with closure of SOEs in the late 1990s.

To explain firm exit, roughly the same explanatory variables that were used in the OLS model for job losses were used in a binary logistic regression model.[9] The results are shown in Table 6. This model estimates 92.4% of cases of firm exit among 681 firms with available data, compared to 62% without these variables in the model.

The model explaining firm exit reinforces the importance of regional variation. All else equal, the location of a firm outside Jiangsu considerably lowered the likelihood of firm exit. In the case of Shanxi, the coefficient is significant and suggests that SOEs in Shanxi faced a 66% lower probability of firm exit. This finding is consistent with the discussion of the provincial variables in explaining changes in firm-level employment. The combined results of the job loss and firm exit models suggest that SOEs in Jiangsu were more likely to exit production, while the SOEs in Jiangsu that survived were in a position to increase labor forces, in comparison to SOEs in other provinces. In Shanxi, Jilin, and Sichuan, by contrast, SOEs were far less likely to exit, and those that remained in production experienced slight declines in payrolls during the late 1990s. This result suggests that the more competitive environment in Jiangsu also provided increased opportunities for employment outside of the state sector.

The firm exit model shows that SOE employment levels in 1994 were positively associated with firm exit, but output value in 1994 was negatively related to firm exit. This result stands in contrast to the job loss model for SOEs, where 1994 employment and 1994 output were positively associated with increases in firm labor in the late 1990s. The seeming discrepancy can be reconciled by considering the taxation variable, which is negative in the firm exit model. SOEs that contributed more taxes were less likely to have exited production; SOEs that paid

[9] Some variables such as percentage of women workers and temporary workers were not available; the percentage of production workers was used. The factory director survey administered in the 1995 survey did not contain questions regarding the extent of Communist Party influence on appointments and had to be dropped in the latter model.

Table 6
Likelihood of firm exit, 1994–1999 (binary logistic regression)

	B	S.E.	Sig.	Exp(B)
Constant	**60.069**	7.979	0.000	n/a
Log 1994 employment	0.593	0.875	0.498	1.810
Log 1994 output value	**−8.314**	0.982	0.000	0.000
% Production workers, 1994	**0.032**	0.018	0.071	1.033
Log 1994 average wage	−0.557	1.413	0.693	0.573
Hiring autonomy	−0.070	0.159	0.661	0.932
Level of reg. agency	0.093	0.255	0.715	1.098
% Retirees	**0.138**	0.051	0.007	1.148
Log 1994 taxes	−0.325	0.302	0.282	0.723
Mining dummy	−0.193	0.808	0.811	0.824
Machinery dummy	−0.214	0.545	0.695	0.808
Utilities dummy	−19.481	12,987.071	0.999	0.000
Food dummy	−0.623	0.741	0.401	0.537
Chemical dummy	−0.838	0.606	0.167	0.433
Metals dummy	−0.567	0.649	0.382	0.567
Furniture–paper dummy	−1.169	0.750	0.119	0.311
Sichuan dummy	−0.045	0.587	0.939	0.956
Shanxi dummy	**−1.061**	0.604	0.079	0.346
Jilin dummy	−0.251	0.560	0.654	0.778

Coefficients for significant variables presented in boldface. $n = 681$.
Percentage correctly predicted = 92.4%.
−2 log likelihood = 269.824.
Nagelkerke R square = 0.769.

none or few taxes (because they were unprofitable) in 1994 were more likely to have ceased production in the late 1990s. After controlling for size of the firm based on employees, firms with both higher output value and higher taxes in 1994 were less likely to exit in the late 1990s. Consistent with central government policy, the larger firms were retained while smaller firms "released," but the models suggest that policymakers were conceiving of scale in terms of taxes, not merely in terms of the number of employees.

The results of the firm exit model are consistent with the industrial sector results in the job loss model. Moving from textiles to generally more capital-intensive sectors lowered the likelihood of firm exit. (Although the furniture and paper firms were considerably less likely to exit than were textiles.) This finding is also consistent with the findings on enterprise scale and tax remissions. The textile sector SOEs, facing heightened competition, were producing goods that sold at lower prices and were no longer capable of remitting taxes. They thus faced a high likelihood of liquidation in the late 1990s.

The proportion of retirees to the total workforce was also a significant explanatory variable for firm exit. Having a large number of retirees to manage (relative to workers) in 1994 meant that a firm had a 14.8% increase in its odds of firm exit. The coefficient for this variable is much stronger than in the models in Table 5 explaining job loss, where the variables were also significant but with very small coefficients. The association of large numbers of retirees and firm exit is plausible given important changes in pension policy during the late 1990s. As the costs for distributing pensions shifted from enterprises to local governments, the latter had a stronger incentive to permit firm bankruptcies and other forms of firm exit by selling off enterprise assets, which could then distributed to retirees under China's bankruptcy regulations. Prior to these reforms, local governments could inject working capital into SOEs by encouraging banks to

extend loans to them, and these loans could be used by the recipient firm to distribute wages and pension benefits.

5. Conclusions

This paper has addressed an apparent paradox of state sector workforce reduction in China in the 1990s, which reveals on the one hand obvious signs of massive reductions in employment in the state sector, and on the other firm-level data that suggest stagnant employment levels or only small declines associated with normal workforce attrition. While it is true that many of the job losses in SOEs are hidden by the fact that SOEs included in employment numbers workers who were no longer paid by the firm or even employed there (except for receiving benefits), a substantial decline in state sector employment also occurred through the exit of firms from the state sector—by ceasing of operations, or by converting to shareholding or other corporate entities.

The preceding discussion of the regression results lends support to variations on each of the hypothesized sources of job destruction—ownership form, industrial sector, and politically driven explanations. State firms in the sample were not adopting flexible labor practices found in foreign-invested and private sectors, as the ownership convergence hypothesis would have predicted. SOEs with lower levels of output and tax remissions exited production by the late 1990s. We cannot know from the 1995 sample of SOEs that would cease production in the late 1990s whether they attempted to cut their workforces before they finally were sold off. We do know from the surviving SOEs that these firms did not cut their payrolls substantially during the late 1990s. The results generally point to a divergence in ownership forms as a significant influence on firm-level labor flexibility.

However, the distinctions in ownership are not based simply on state versus "non-state" sectors. Both of these categories contain multiple forms of ownership. Traditional state enterprises, collectives, and shareholding (or corporatized) state firms are subject to intervention and interference by various state agents who might claim partial property rights over particular groups of firms. The regression models showed that different variables were significant across ownership categories, and even operated in different causal directions across them. Urban collectives responded to an increasingly competitive environment by shedding labor. Traditional state enterprises that did not transform into shareholding entities, however, reported steady or far smaller declines in employment levels. Greater levels of workforce reduction among SOEs were found in labor-intensive and competitive sectors such as textiles. Firm exit among SOEs was likely to take place among enterprises with lower output values; after controlling for scale, higher proportions of retirees increased the likelihood of firm exit.

Job destruction in China has remained politically driven as well. Variables such as the administrative level of the agency supervising the firm, and the degree of Communist Party influence on appointments in a firm, proved to be insignificant in the regression models. Region, however, was an important variable in explaining both firm-level employment changes and firm exit. The fact that state enterprises in capital-intensive sectors and in interior provinces had lower likelihoods of exiting production provides strong evidence of a strategic approach to state sector restructuring. Larger SOEs in more capital-intensive sectors that remitted higher levels of taxes were likely to survive 1990s restructuring, whereas smaller, labor-intensive SOEs were not. Surviving SOEs in interior provinces reported roughly the same if not larger workforce levels in the late 1990s.

However, the political constraints hypothesis for job destruction has important limits. The argument that the CCP regime has avoided massive layoffs out of fear of political instability is

not valid, given that state sector employment, in manufacturing in particular, declined so rapidly in the late 1990s. To the extent that there was a coordinated political strategy to job destruction in the state sector, it appeared to be one that relied upon a "commanding heights" approach, in which large firms in strategic sectors, along with their workers, were retained by the state. For those firms and workers outside the commanding heights, in collective, shareholding, private, and foreign-invested enterprises, job destruction and creation are likely to follow the logic of the marketplace.

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